

Digital Logic Design

**Semester II
FY**

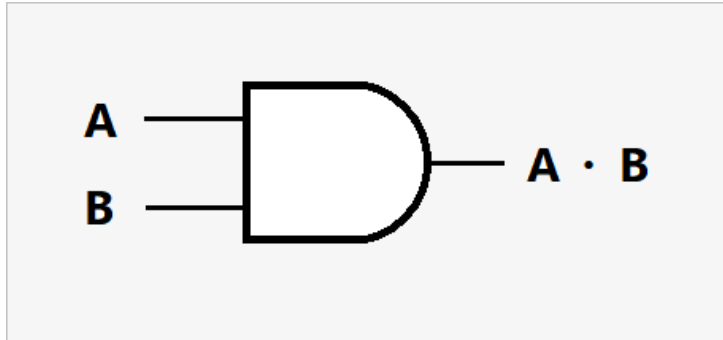
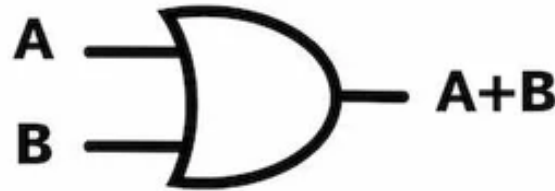
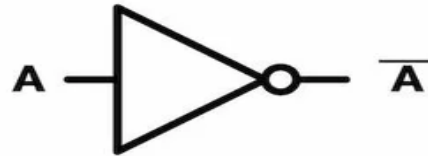
**Course Code
316U06C203
Jan – May 2026**

Logic Representation and Minimization Techniques

Contents

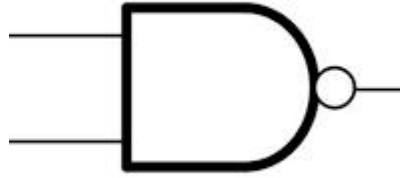
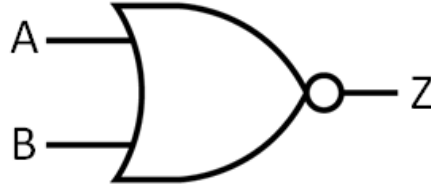
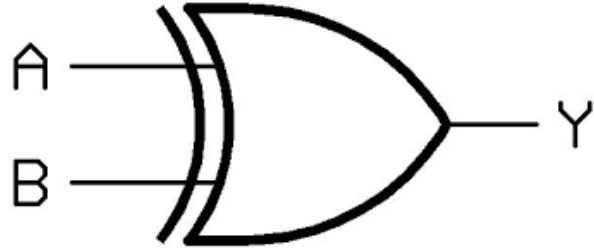
- **Basic gates AND, OR, NOT**
- **Derived gates : NAND, NOR, XOR,**
- **Boolean Theorems**
- **Representation of Logic in Boolean Equation**
- **SOP and POS form**
- **Minimization of Boolean equation using Algebra**
- **Minimization using K Map**

Basic Gates

Gate Name	Symbol	Truth Table															
AND		<table><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Y	0	0	0	0	1	0	1	0	0	1	1	1
A	B	Y															
0	0	0															
0	1	0															
1	0	0															
1	1	1															
OR		<table><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Y	0	0	0	0	1	1	1	0	1	1	1	1
A	B	Y															
0	0	0															
0	1	1															
1	0	1															
1	1	1															
NOT		<table><tr><th>A</th><th>A'</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	A'	0	1	1	0									
A	A'																
0	1																
1	0																

MAIYA

Derived Gates

Gate Name	Symbol	Truth Table															
NAND		<table> <tr> <th>A</th><th>B</th><th>Y</th></tr> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </table>	A	B	Y	0	0	1	0	1	1	1	0	1	1	1	0
A	B	Y															
0	0	1															
0	1	1															
1	0	1															
1	1	0															
NOR		<table> <tr> <th>A</th><th>B</th><th>Y</th></tr> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </table>	A	B	Y	0	0	1	0	1	0	1	0	0	1	1	0
A	B	Y															
0	0	1															
0	1	0															
1	0	0															
1	1	0															
XOR		<table> <tr> <th>A</th><th>B</th><th>Y</th></tr> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </table>	A	B	Y	0	0	0	0	1	1	1	0	1	1	1	0
A	B	Y															
0	0	0															
0	1	1															
1	0	1															
1	1	0															

Boolean Algebra Theorems

1. Identity Laws

$$A + 0 = A$$

$$A \cdot 1 = A$$

2. Null/Annihilation Laws

$$A + 1 = 1$$

$$A \cdot 0 = 0$$

3. Idempotent Laws

$$A + A = A$$

$$A \cdot A = A$$

4. Complement Laws

$$A + A' = 1$$

$$A \cdot A' = 0$$

5. Double Negation Law

$$(A')' = A$$

6. Commutative Laws

$$A + B = B + A$$

$$A \cdot B = B \cdot A$$

Boolean Algebra Theorems..

7. Associative Laws

$$(A + B) + C = A + (B + C)$$

$$(A \cdot B) \cdot C = A \cdot (B \cdot C)$$

8. Distributive Laws

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

9. Absorption Laws

$$A + A \cdot B = A$$

$$A \cdot (A + B) = A$$

10. DeMorgan's Theorems

$$(A \cdot B)' = A' + B'$$

$$(A + B)' = A' \cdot B'$$

11. Redundancy Laws

$$A + A' \cdot B = A + B$$

$$A \cdot (A' + B) = A \cdot B$$

Simplification using Boolean Theorems

Given an Boolean Expression check whether it is optimized

$$Y = AB + AB'C + C' + ABC$$

Circuit requires **2 i/p AND 5 no, NOT 2 No, 2 i/p OR 3 No**

$$\begin{aligned} Y &= AB + AB'C + C' + \textcolor{red}{ABC} + \textcolor{red}{ABC} \dots\dots A + A = A \\ &= AB + ABC + AB'C + C' + ABC \dots \text{commutative} \\ &= \textcolor{red}{AB} (1 + C) + AB'C + C' + ABC \dots \text{Distributive} \\ &= AB + AB'C + ABC + C' \dots 1 + A = 1 \\ &= AB + \textcolor{red}{AC(B' + B)} + C' \dots \text{Distributive} \\ &= AB + \textcolor{red}{AC} + C' \\ &= \textcolor{red}{AB + A + C'} = A(B + 1) + C' = A + C' \end{aligned}$$

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$Y = A'B'C' + A'BC' + AB'C' + AB'C + ABC' + ABC$$

Simplify

$$F(A, B, C, D) = (A + B').(ACB + B'.[C + A]) + ABC'.(A'B + A)$$

Identify how many inputs...D not required

Always maintain their sequence (even if it does not matter)

$$\begin{aligned} F &= (A+B') . (ABC+B'C+AB') + (AA'BBC' + AABC') \\ &= (AABC+AB'C+AAB'+ABB'C+B'B'C+AB'B') + (AA'BBC' + AABC') \\ &= ABC+AB'C+AB'+0+B'C+AB' +0+ABC' \\ &= ABC+AB'(C+1)+B'C+AB'+ABC' \\ &= AB(C+C') +AB' +B'C \\ &= AB+ AB'+B'C \\ &= A(B+B') + B'C \\ &= A+B'C \end{aligned}$$

Representation of Logic using Boolean Expression

Write a truth table for function which is one when the four digit number is prime number

No of Inputs?

Truth Table

Minterms

Maxterms

Output expressed as a function of inputs

= sum of minterms = sum of Products = SOP

= Product of Maxterms = Product of Sums

Boolean Expression

$$\begin{aligned} P &= A'B'CD' + A'B'CD + A'BC'D + A'BCD + AB'CD + ABC'D \\ &= A'B'C + A'BD + AB'CD + ABC'D \end{aligned}$$

A	B	C	D	Prime
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

Prime = 1 when input is 2,3,5,7,11,13

- Choose rows of truth table having output 1
- Look at inputs in that row
- Write an expression for the row as product (AND)
- If variable is having value zero write its complement, if it is having value 1 write the variable as it is

Representation of Logic in SOP and POS form

- In any Truth Table Output Values are either zero / one
- **Minterms** are used to represent the logic expression for **output = 1**.
- A minterm is a **product (AND)** of all input variables in either true or complemented form, such that only one combination results in 1.
- **Maxterms** are used to represent the logic expression for **output = 0**.
- A maxterm is a **sum (OR)** of all input variables in either true or complemented form, such that only one combination results in 0.

Representation of Logic in SOP and POS form

Any Truth Table can be represented as

- Sum of Products (Minterms)
- Product of Sums(Maxterms)
- $P = m_2 + m_3 + m_5 + m_7 + m_{11} + m_{13}$ (SOP)
- $P = M_0 \cdot M_1 \cdot M_4 \cdot M_6 \cdot M_8 \cdot M_9 \cdot M_{10} \cdot M_{12} \cdot M_{14} \cdot M_{15}$ (POS)
- Minterms are represented as Products
- $m_2 = A'B'CD'$, $m_3 = A'B'CD$, $m_{11} = AB'CD$ etc
- Maxterms are represented as Sums
- $M_0 = A+B+C+D$, $M_1 = A+B+C+D'$, $M_6 = A+B'+C'+D$
- $M_{12} = A'+B'+C+D$, $M_{15} = A'+B'+C'+D'$ etc

Representation of Logic in SOP and POS form

SOP form of boolean expression can also be written as

$$P = \sum m(2,3,5,7,11,13) \dots \text{SOP}$$

POS form as

$$P = \prod M(0,1,4,6,8,9,10,12,14,15) \dots \text{POS}$$

Total terms (rows of the truth table) are dependent on number of inputs

For n inputs terms = 2^n

Representation of Logic in SOP and POS form

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Total terms (rows of the truth table) are dependent on number of inputs

For n inputs terms = 2^n

SOP (Sum of Products)

Minterms (Output = 1)

OR of ANDs (e.g., $A \cdot B + C$)

Logic implementation via AND-OR networks

POS (Product of Sums)

Maxterms (Output = 0)

AND of ORs (e.g., $(A + B) \cdot (C + D)$)

Logic implementation via OR-AND networks

Representation of Logic in SOP and POS form

$$\begin{aligned} \text{Consider function } Y(A,B,C,D) &= \sum m(2,3,5,7,8,10,11,13) \dots \text{SOP} \\ Y &= A'B'CD' + A'B'CD + A'BC'D + A'BCD + AB'C'D' + AB'CD' + AB'CD + ABC'D \\ &= A'B'CD' + A'B'CD + A'BC'D + A'BCD + AB'C'D' + AB'CD' + AB'CD + ABC'D \\ &= A'B'C(D' + D) + A'BD(C' + C) + AB'D'(C' + C) + AB'CD + ABC'D \\ &= A'B'C + A'BD + AB'D + AB'CD + ABC'D \\ &= B'C(A' + AD) + A'BD + AD(B' + BC') \dots\dots\dots A' + AD = A' + D, B' + BC' = B' + C' \\ &= B'C(A' + D) + A'BD + AD(B' + C') \\ &= A'B'C + B'CD + A'BD + AB'D + AC'D \end{aligned}$$

Representation of Expression in Karnaugh Maps

Consider function $Y(A,B,C) = \sum m(1,2,3,4,7) \dots \text{SOP}$

$$\begin{aligned} Y &= A'B'C + A'BC' + A'BC + AB'C' + ABC = A'C(B' + B) + A'B(C + C') + BC(A' + A) + AB'C' \\ &= A'C + A'B + BC + AB'C' \end{aligned}$$

AB/C	00	01	11	10
0	0	1	0	1
1	1	1	1	0

$$A'C + A'B + BC + AB'C'$$

Minimization using Karnaugh Maps

Consider function $Y(A,B,C,D) = \sum m(1,2,3,4,7,8,11,13,15) \dots \text{SOP}$

AB/CD	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	1	1	1	1
10	1	0	0	0

$$A'B'D + A'B'C + CD + ABD + A'BC'D' + AB'C'D'$$

Karnaugh Maps Minimization Rules

1. Identify all minterms in the given function. (or maxterms)
2. Group adjacent minterms for SOP (maxterms for POS)
3. Groups having 1,2,4,8...upto 2^n for n variable K Map
4. K-maps are cylindrical: the left and right edges are adjacent, as are the top and bottom. This allows wrap-around groupings.
5. A cell can be part of multiple groups if it helps make larger groups.
6. Make Largest Possible Groups First
7. All minterms must be covered even if it is a group of 1 term
8. Make sure that no group is redundant (redundant group is the one in which all minterms are part of some other group)
9. For each group:
Eliminate the variables that change within the group.
Keep variables that are the same for all cells in the group.
10. To represent each group consider a product with variable (for 1) and its complement (for 0) for SOP

Minimization using Karnaugh Maps

Consider function $Y(A,B,C,D) = \sum m(0,1,2,3,4,6,7,8,11,13,15) \dots \text{SOP}$

AB/CD	00	01	11	10
00	1	1	0	1
01	1	0	1	0
11	1	1	1	1
10	1	1	0	0

$$A'B' + A'D' + CD + ABD + B'C'D'$$

Karnaugh Maps with Don't care terms:

Consider function $Y(A,B,C,D) = \sum m(0,1,2,4,6,11,15) + d(3,9,13) \dots \text{SOP}$

AB/CD	00	01	11	10
00	1	1	0	0
01	1	0	X	X
11	X	0	1	1
10	1	1	0	0

$Y = A'D' + A'B'C' + ACD \dots$ without considering don't care terms

AB/CD	00	01	11	10
00	1	1	0	0
01	1	0	X	X
11	X	0	1	1
10	1	1	0	0

$Y = A'D' + A'B' + AD \dots$ with Don't cares

Minimization using Karnaugh Maps (POS)

Consider function $Y(A,B,C,D) = \sum m(0,1,2,3,4,6,7,8,11,13,15) \dots \text{SOP}$
 $= \prod M(5,9,10,12,14) \dots \text{POS}$

AB/CD	00	01	11	10
00	1	1	0	1
01	1	0	1	0
11	1	1	1	1
10	1	1	0	0

$$Y = (A+B'+C+D')(A'+B+C+D')(A'+B'+D)(A'+C'+D)$$

$$A'B' + A'D' + CD + ABD + B'C'D'$$

Minimization using Karnaugh Maps

1. $F(A,B,C,D) = \sum m(0, 2, 4, 5, 8, 10, 12, 14, 15)$
2. $F(A,B,C,D) = \sum m(0,1,4,5,10,11,13,14,15)$
3. $F(A,B,C,D) = \prod M(0,1,4,5,10,11,13,14,15)$
4. $F(A,B,C,D) = \prod M(0,1,6,7,9,11,12,13,15)$

AB/CD	00	01	11	10
00				
01				
11				
10				